Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with experience building high-performance, **multi-threaded** desktop and client applications in **C#**, **WinForms**, and **.NET**. Strong in real-time data visualization, OOP, unit testing, and collaborating with technical and business partners in fast-paced environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, OOP, Operating Systems, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Optimized real-time dashboard performance from **1.2s to 350ms** through indexing and concurrent job orchestration, building visualization-ready data services for **500+ users** across production cloud and SQL-backed workflows.
- Collaborated with stakeholders to define requirements, ship tested releases, and troubleshoot production issues while maintaining **99%+ reliability** across **5K+ monthly integration jobs**.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed concurrent processing pipelines and operator-facing workflow tools handling **10K+ daily requests**, partnering with end users to translate feedback into reliable feature enhancements under changing priorities.
- Implemented performance-tuned backend services with structured logging and automated test coverage, supporting high-throughput review interfaces and reducing manual turnaround by **60%** in production.

Aasma Technology Solutions **India**
Software Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered high-performance **Windows** client integrations and service modules with REST APIs, improving response times by **30%** through caching, pagination, and OOP refactoring validated by **20+ unit test suites**.
- Maintained cross-platform desktop-adjacent delivery workflows in Agile teams, communicating technical tradeoffs clearly while supporting multiple concurrent initiatives with Microsoft/GitHub business partners.

Aasma Technology Solutions **India**
Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*

- Built real-time operational dashboards with WebSocket-driven updates across **2M+ records**, creating custom visualization modules and scalable UI data flows for distributed client operations teams.
- Automated testing and deployment pipelines with Jenkins and Docker, improving release reliability while adapting quickly to changing feature priorities across billing, resource, and reporting desktop-style workflows.

Enterprise Applications Lab **India**
Desktop Frontend Software Engineer Intern *November 2022 – January 2023*

- Built **multi-threaded C#/WinForms** desktop modules with **Syncfusion** controls for real-time operational dashboards, delivering responsive UI updates and internal reporting workflows used by business analysts on **Windows**.
- Applied OOP design patterns and **unit tests** while profiling memory and UI latency with .NET diagnostics, improving form rendering performance and maintainability across concurrent data refresh scenarios.

Skills

Desktop & .NET: C#, WinForms, WPF (familiar), .NET, Core .NET, multi-threading, high-performance UI, DevExpress (familiar), Telerik (familiar), Syncfusion (familiar), OOP, unit testing, memory profiling (familiar), performance profiling (familiar), Windows
Engineering: real-time data visualization, desktop applications, concurrent programming, async patterns, REST APIs, SQL Server, Git, CI/CD, Agile, problem solving, communication, trading systems (familiar), market structure (familiar), Python, Java

Projects

TradeDesk UI Code | *C#, WinForms, Multi-threading, Real-Time Visualization*

- Built a desktop **C#/WinForms** prototype for streaming market-style data grids with **multi-threaded** refresh workers, custom charting views, and unit-tested UI components for low-latency visualization.

CPU-scheduling Simulator Live | *C++, Concurrency, Performance Engineering*

- Implemented concurrent scheduling algorithms in **C++** with automated validation across **1K+ process combinations**, reducing debugging time by **20%** through performance-focused systems engineering.

FinNexus Live | *Real-Time Data, High-Volume Transactions, Filtering*

- Developed a fintech platform with real-time transaction updates and advanced filtering across **1M+ records**, demonstrating high-throughput operational UI patterns relevant to live data monitoring workflows.